Assignment No : 03- Descriptive Statistics - Measures of Central Tendency and variability

Problem Statement:

Perform the following operations on any open source dataset (e.g., data.csv)

1. Provide summary statistics (mean, median, minimum, maximum, standard deviation) for a dataset (age, income etc.) with numeric variables grouped by one of the qualitative(categorical) variable. For example, if your categorical variable is age groups and quantitative variable is income, then provide summary statistics of income grouped by the age groups. Create a list that contains a numeric value for each response to the categorical variable.
2. Write a Python program to display some basic statistical details like percentile, mean, standard deviation etc. of the species of 'Iris-setosa', 'Iris-versicolor' and 'Iris-versicolor' of iris.csv dataset. Provide the codes with outputs and explain everything that you do in this step.

Step 1 : Import Required Libraries

```
import pandas as pd
import numpy as np
```

Step 2 : Load the Dataset

```
df = pd.read_csv('/content/Iris.csv')
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
						Iris-

Next steps: Generate code with df New interactive sheet

Step 3 : Check Dataset Information

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
df.columns
```

```
Index(['Id', 'SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm',
       'Species'],
      dtype='object')
```

```
df.size
```

```
900
```

```
df.shape
```

```
(150, 6)
```

Step 4: Summary Statistics

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

Step 5: Group by Categorical Variable (Species)

```
grouped_stats = df.groupby("Species").agg(['mean','median','min','max','std'])
grouped_stats
```

	Id					SepalLengthCm					..
	mean	median	min	max	std	mean	median	min	max	std	..
Species											
Iris-setosa	25.5	25.5	1	50	14.57738	5.006	5.0	4.3	5.8	0.352490	
Iris-versicolor	75.5	75.5	51	100	14.57738	5.936	5.9	4.9	7.0	0.516171	
Iris-virginica	125.5	125.5	101	150	14.57738	6.588	6.5	4.9	7.9	0.635880	

3 rows × 25 columns

Step 6: Create List for Each Category

```
setosa_list = df[df['Species']=="Iris-setosa"]['PetalLengthCm'].tolist()
versicolor_list = df[df['Species']=="Iris-versicolor"]['PetalLengthCm'].tolist(
virginica_list = df[df['Species']=="Iris-virginica"]['PetalLengthCm'].tolist()

print("Setosa:", setosa_list[:5])
print("Versicolor:", versicolor_list[:5])
print("Virginica:", virginica_list[:5])
```

```
Setosa: [1.4, 1.4, 1.3, 1.5, 1.4]
Versicolor: [4.7, 4.5, 4.9, 4.0, 4.6]
Virginica: [6.0, 5.1, 5.9, 5.6, 5.8]
```

Step 7: Statistical Details for Each Species

```
setosa = df[df['Species']=="Iris-setosa"]
versicolor = df[df['Species']=="Iris-versicolor"]
virginica = df[df['Species']=="Iris-virginica"]

print("Setosa Statistics")
print(setosa.describe())

print("\nVersicolor Statistics")
print(versicolor.describe())

print("\nVirginica Statistics")
print(virginica.describe())
```

```
Setosa Statistics
             Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
count  50.00000       50.00000     50.000000      50.000000      50.00000
mean   25.50000        5.00600      3.418000       1.464000       0.24400
std    14.57738        0.35249      0.381024       0.173511       0.10721
min     1.00000        4.30000      2.300000       1.000000       0.10000
25%    13.25000        4.80000      3.125000       1.400000       0.20000
50%    25.50000        5.00000      3.400000       1.500000       0.20000
75%    37.75000        5.20000      3.675000       1.575000       0.30000
max    50.00000        5.80000      4.400000       1.900000       0.60000

Versicolor Statistics
              Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
count   50.00000      50.000000     50.000000      50.000000     50.000000
mean    75.50000       5.936000      2.770000       4.260000      1.326000
std     14.57738       0.516171      0.313798       0.469911      0.197753
min     51.00000       4.900000      2.000000       3.000000      1.000000
25%     63.25000       5.600000      2.525000       4.000000      1.200000
50%     75.50000       5.900000      2.800000       4.350000      1.300000
75%     87.75000       6.300000      3.000000       4.600000      1.500000
max    100.00000       7.000000      3.400000       5.100000      1.800000

Virginica Statistics
              Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  PetalWidthCm
count   50.00000       50.00000     50.000000      50.000000      50.00000
mean   125.50000        6.58800      2.974000       5.552000       2.02600
std     14.57738        0.63588      0.322497       0.551895       0.27465
min    101.00000        4.90000      2.200000       4.500000       1.40000
25%    113.25000        6.22500      2.800000       5.100000       1.80000
50%    125.50000        6.50000      3.000000       5.550000       2.00000
75%    137.75000        6.90000      3.175000       5.875000       2.30000
max    150.00000        7.90000      3.800000       6.900000       2.50000
```

Step 8: Percentile Calculation

```
print("25th Percentile")
print(df.groupby('Species').quantile(0.25))

print("\n50th Percentile")
print(df.groupby('Species').quantile(0.50))

print("\n75th Percentile")
print(df.groupby('Species').quantile(0.75))
```

```
25th Percentile
                     Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  \
Species
Iris-setosa       13.25          4.800         3.125            1.4
Iris-versicolor   63.25          5.600         2.525            4.0
Iris-virginica   113.25          6.225         2.800            5.1

                 PetalWidthCm
Species
Iris-setosa               0.2
Iris-versicolor           1.2
Iris-virginica            1.8

50th Percentile
                    Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  \
Species
Iris-setosa       25.5            5.0           3.4           1.50
Iris-versicolor   75.5            5.9           2.8           4.35
Iris-virginica   125.5            6.5           3.0           5.55

                 PetalWidthCm
Species
Iris-setosa               0.2
Iris-versicolor           1.3
Iris-virginica            2.0

75th Percentile
                     Id  SepalLengthCm  SepalWidthCm  PetalLengthCm  \
Species
Iris-setosa       37.75            5.2         3.675          1.575
Iris-versicolor   87.75            6.3         3.000          4.600
Iris-virginica   137.75            6.9         3.175          5.875

                 PetalWidthCm
Species
Iris-setosa               0.3
Iris-versicolor           1.5
Iris-virginica            2.3
```

Conclusion

In this practical, descriptive statistical techniques such as mean, median, minimum, maximum, standard deviation, and percentiles were calculated using Python. The Iris Dataset was analyzed by grouping the numeric variables based on the categorical variable Species to understand the distribution of different flower species. This analysis helped in identifying the central tendency and variability of the dataset, demonstrating how descriptive statistics can be used to summarize and interpret data effectively.